➢ M-ARY PSK and QAM

The general equation of M-ARY **PSK (MPSK)** signals in the interval t = [0, T_b] can be written as

$$\varphi_{MPSK} = \sqrt{\frac{2E}{T_b}} \cos\left[2\pi f_c + (m-1)\frac{2\pi}{M}\right] \qquad m = 1,2,\dots,M$$

$$= \sqrt{\frac{2E}{T_b}} \cos\left[(m-1)\frac{2\pi}{M}\right] \cos(2\pi f_c) - \sqrt{\frac{2E}{T_b}} \sin\left[(m-1)\frac{2\pi}{M}\right] \sin(2\pi f_c)$$

$$= \sqrt{E} \cos\left[(m-1)\frac{2\pi}{M}\right] \psi_1(t) - \sqrt{E} \sin\left[(m-1)\frac{2\pi}{M}\right] \psi_2(t)$$

where

$$\psi_1(t) = \sqrt{\frac{2}{T_b}} cos\,(2\pi f_c)\,, \text{ and } \qquad \psi_2(t) = \sqrt{\frac{2}{T_b}} sin\,(2\pi f_c)$$

- It can be seen that $\psi_1(t)$ and $\psi_2(t)$ are basis functions because they orthogonal to each other. In addition, they are normalized over $[0, T_b]$ provided that $f_c T_b$ = integer. As a result, we can represent all PSK symbols in a two-dimensional signal space using these basis functions.
- The MPSK signal can be represented as

$$\varphi_{MPSK} = A cos\theta_m \psi_1(t) - A sin\theta_m \psi_2(t) \quad \text{(A)}$$

where $\theta_m = (m-1)\frac{2\pi}{M}$ and $A = \sqrt{E}$

Let $a_m = A cos\theta_m$, $b_m = -A sin\theta_m$

$$\therefore \varphi_{MPSK} = a_m \psi_1(t) + b_m \psi_2(t) \quad \text{(B)}$$

- We can geometrically illustrate the relationship of the **PSK** symbols in the signal space (Fig. 7.37). These geometric representation are called the **constellation diagrams.**

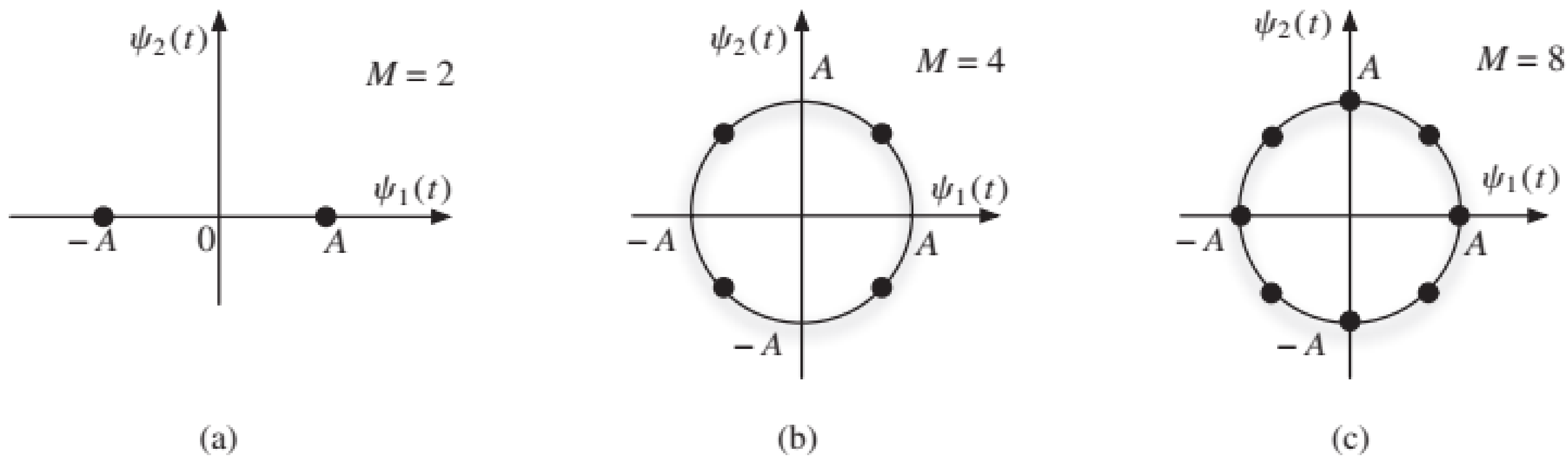


Figure 7.37: The constellation diagram of *M*-ary PSK symbols in the orthogonal signal space: (a) M = 2; (b) M = 4; (c) M = 8.

- Eq. (A) means that PSK modulations can be represented as QAM signal. In fact, because the signal is **PSK,** the signal points must meet a special requirement that

$$
\begin{aligned}
a_m^2 + b_m^2 &= A^2 \cos^2 \theta_m + (-A)^2 \sin^2 \theta_m \\
&= A^2 = \text{constant}
\end{aligned}
$$

- In other words, all the signal points must stay on a circle of radius *A.* In practice, all the signal points are chosen to be equally spaced to obtain the best immunity against noise .
- The special PSK signaling with *M* = 4 is an extremely popular and powerful digital modulation format. It in fact is a summation of two binary PSK signals, one using the (in-phase) carrier of $\cos\omega_c t$ while the other uses the (quadrature) carrier of $\sin\omega_c t$ of the same frequency.
- For this reason, it is also known as ***quadrature PSK (QPSK).*** We can transmit and receive both of these signals on the same channel, thus doubling the transmission rate.

- To further generalize the PSK to achieve higher data rate, we can see that the PSK representation of Eq. (A) is a special case of the quadrature amplitude modulation (QAM) signal discussed in Chapter 4 (Fig. 4. 19). The only difference lies in the requirement by PSK that the modulated signal have a constant magnitude ($A = \sqrt{E}$). In fact, the much more flexible and general QAM signaling format can be conveniently used for digital modulation as well. The signal transmitted by an M-ary QAM system can be written as

$$
\begin{aligned}
p_i(t) &= a_i p(t) \cos \omega_c t + b_i p(t) \sin \omega_c t \\
&= r_i p(t) \cos (\omega_c t - \theta_i) \qquad\qquad i = 1, 2, \ldots, M
\end{aligned}
$$

where

$$
r_i = \sqrt{a_i^2 + b_i^2} \qquad \text{and} \qquad \theta_i = \tan^{-1} \frac{b_i}{a_i}
$$

and p(t) is a properly shaped baseband pulse. The simplest choice of p(t) would be a full-width rectangular pulse

$$p(t) = \sqrt{\frac{2}{T_b}}\,[u(t) - u(t - T_b)]$$

Certainly, better pulses can also be designed to conserve bandwidth.

- Figure 7.38a shows the QAM modulator and demodulator. Each of the two signals $m_1(t)$ and $m_2(t)$ is a baseband M-ary pulse sequence. The two signals are modulated by two carriers of the same frequency but in phase quadrature.
- The i-th digital QAM signal pi(t) can be generated by letting $m_1(t) = a_i p(t)$ and $m_2(t) = b_i p(t)$ in a standard QAM transmitter. Both $m_1(t)$ and $m_2(t)$ are baseband PAM signals.

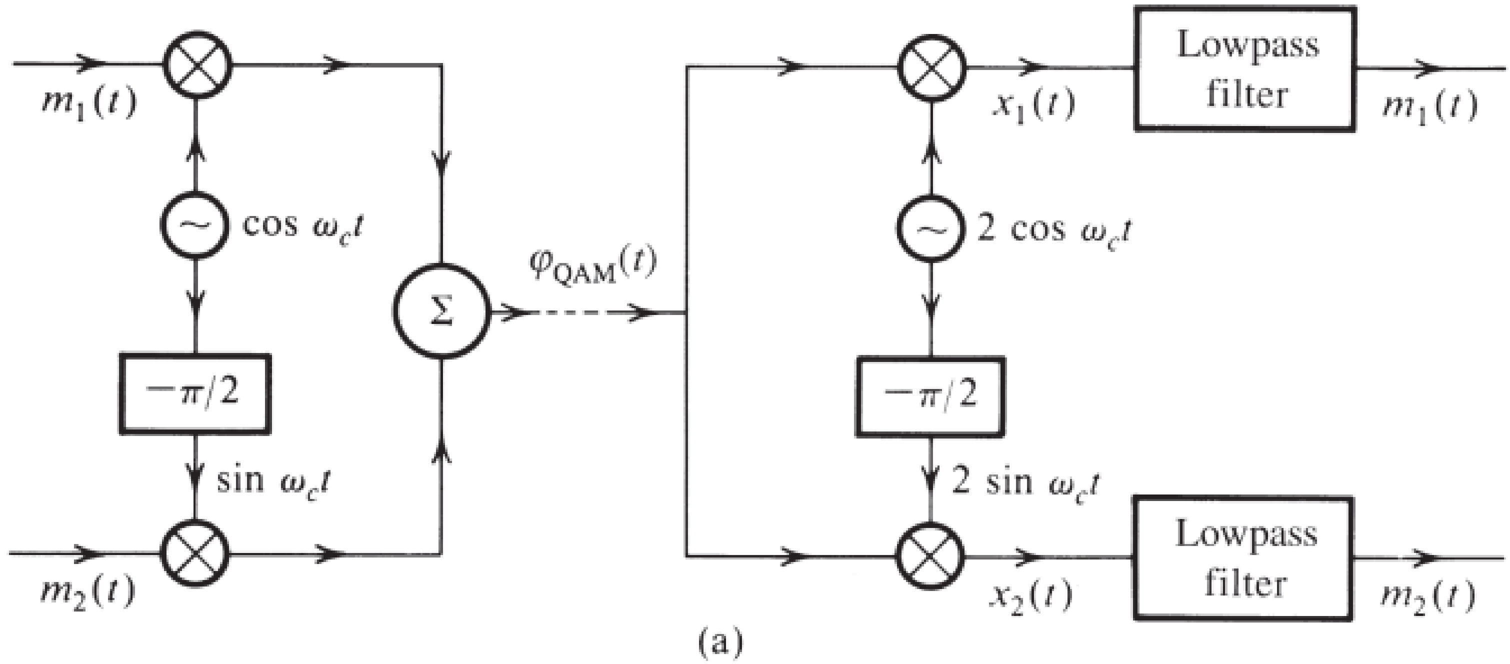
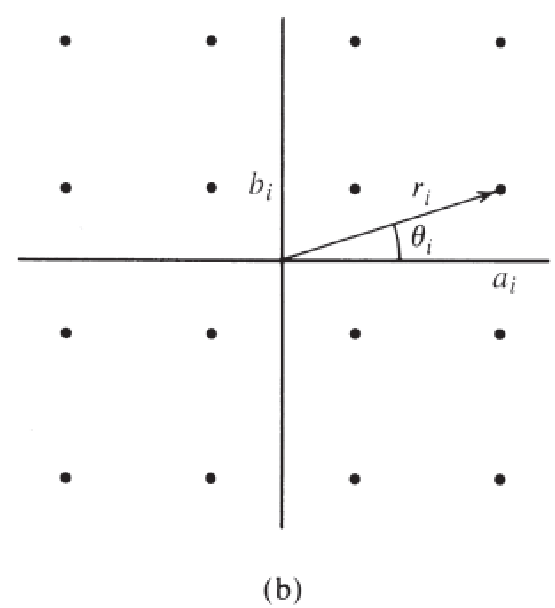


Figure 7.38
(a) QAM or quadrature multiplexing and (b) the constellation diagram of 16-point QAM *(M* = 16*)*.



- The constellation diagram of M-ary QAM can be extended from the PSK signal space by simply removing the constant modulus constraint in Eq. (A).
- One very popular and practical choice of r_i and θ_i for M = 16 is shown graphically in Fig. 7.38b. The transmitted pulse $p_i(t)$ can take on 16 distinct forms, and is, therefore, a 16-ary pulse. Since M = 16, each pulse can transmit the information of $\log_2 16 = 4$ binary digits.
- This can be done as follows: there are 16 possible sequences of four binary digits and there are 16 combinations (a_i, b_i) in Fig. 7.38b. Thus, every possible four-bit sequence is transmitted by a particular (a_i, b_i) or (r_i, θ_i). Therefore, one signal pulse $r_i p(t)\cos(\omega_c t - \theta_i)$ transmits four bits.

- Compared with binary PSK (or BPSK), the 16-ary QAM bit rate is quadrupled without increasing the bandwidth. The transmission rate can be increased further by increasing the value of M.
- The practical value of this 16-ary QAM signaling becomes fully evident when we consider its broad range of applications, including 4G-LTE cellular transmission, DOCSIS (Data Over Cable Service Interface Specification) modem transmissions, DSL broadband service, and satellite digital video broadcasting.
- Note that if we disable the data stream that modulates the carrier $\sin \omega_c t$ in QAM, then all the signaling points can be reduced to a single dimension. Upon setting $m_2(t) = 0$, QAM becomes

$$p_i(t) = a_i p(t) \cos \omega_c t, \qquad t \in [0, T_b]$$

- This degenerates into PAM signaling.
- Comparison of the signal expression of $p_i(t)$ with the analog DSB-SC signal makes it clear that PAM is the digital version of the DSB-SC signal.
- Just as analog QAM is formed by the superposition of two DSB-SC amplitude modulations in phase quadrature, digital QAM consists of two PAM signals, each having $\sqrt{M}$ signaling levels. Similarly, like the relationship between analog DSB-SC and analog QAM, PAM requires the same amount of bandwidth as digital QAM does. However, PAM is much less efficient in both spectrum and power because it would need M modulation signaling levels in one dimension, whereas QAM requires only $\sqrt{M}$ signaling levels in each of the two orthogonal QAM dimensions.

➢ Trading Power for Bandwidth

- The kind of exchange between the transmission bandwidth and the transmitted power (or SNR) depends on the choice of M-ary modulation scheme. For example, in orthogonal signaling, the transmitted power is practically independent of M, but the transmission bandwidth grows with M. Contrast this to the PAM case, where the transmitted power increases roughly with M^2 while the bandwidth remains constant. Thus, M-ary signaling allows greater flexibility in trading signal power (or SNR) for transmission bandwidth. The choice of an appropriate system will depend upon the particular application and circumstance. For instance, it will be appropriate to use QAM signaling if the bandwidth is at a premium (as in DSL lines) and to use orthogonal signaling when power is more critical (as in deep space communication).